

CLM

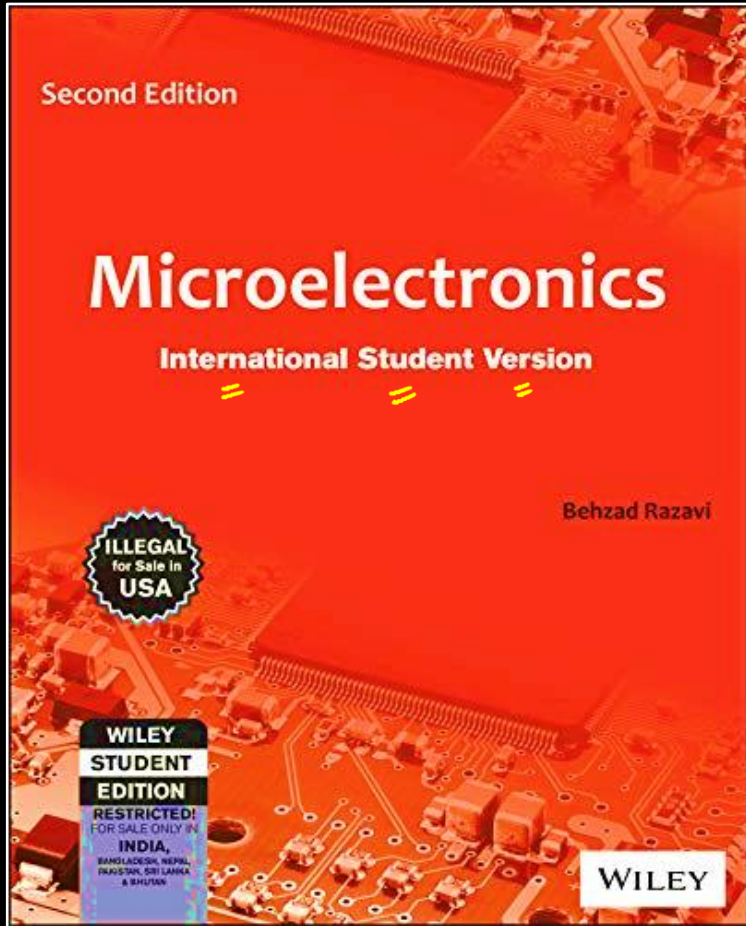
LECTURE – 2 :: (25-03-2025)

Er. Sailesh Mishra

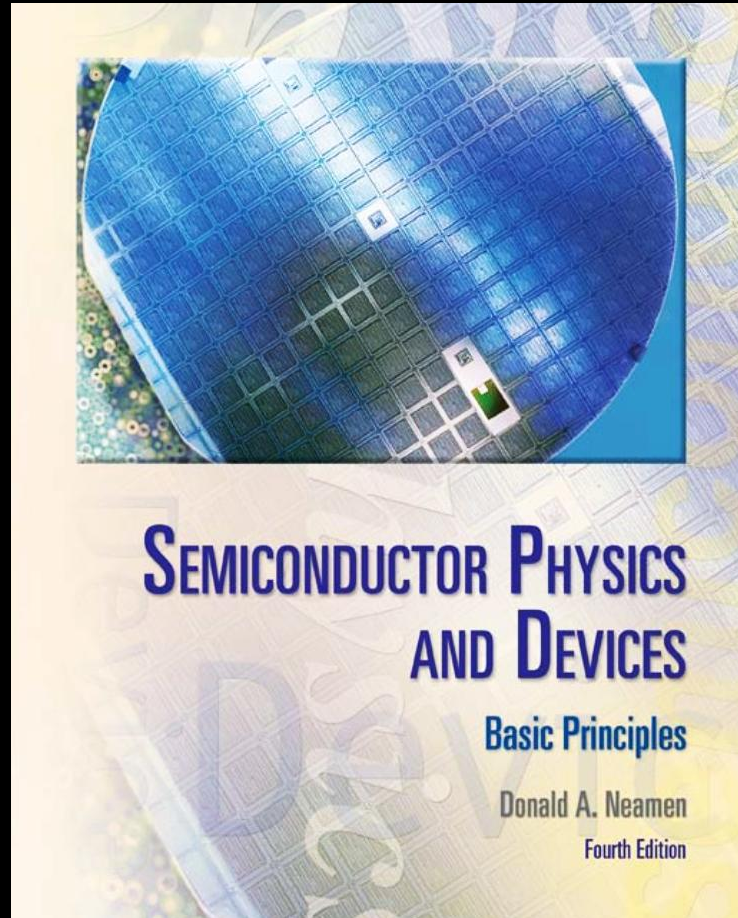
[Spec. Microelectronics & VLSI]

Silicon University, Bhubaneswar

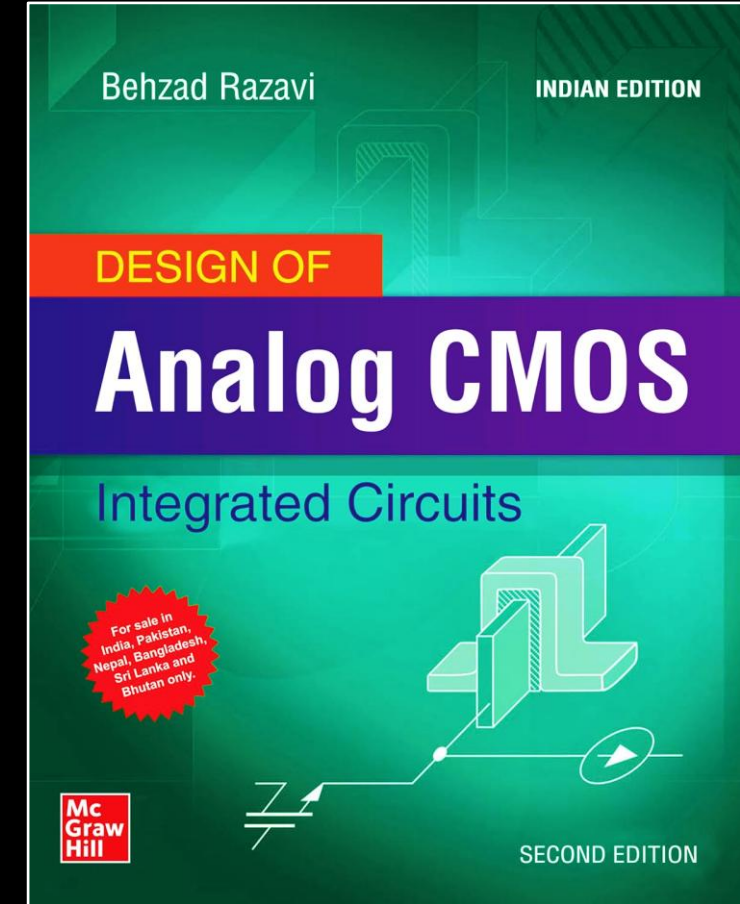
ANALOG VLSI THEORY BOOKS



→ SYNOPSIS, SEVYA, ANALOG DEVICES

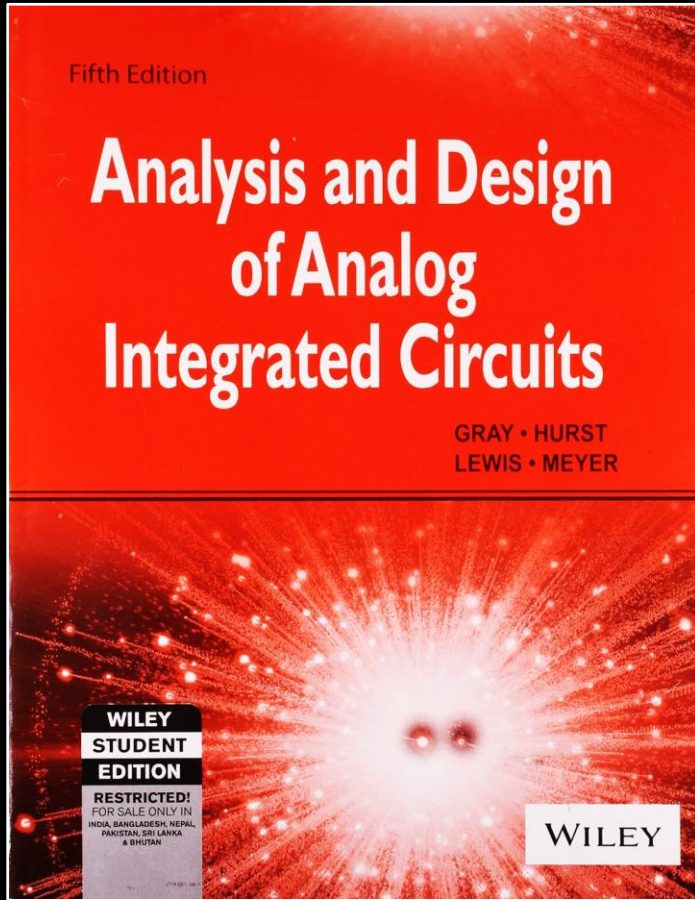


→ p-n junction; DEVICE PHYSICS

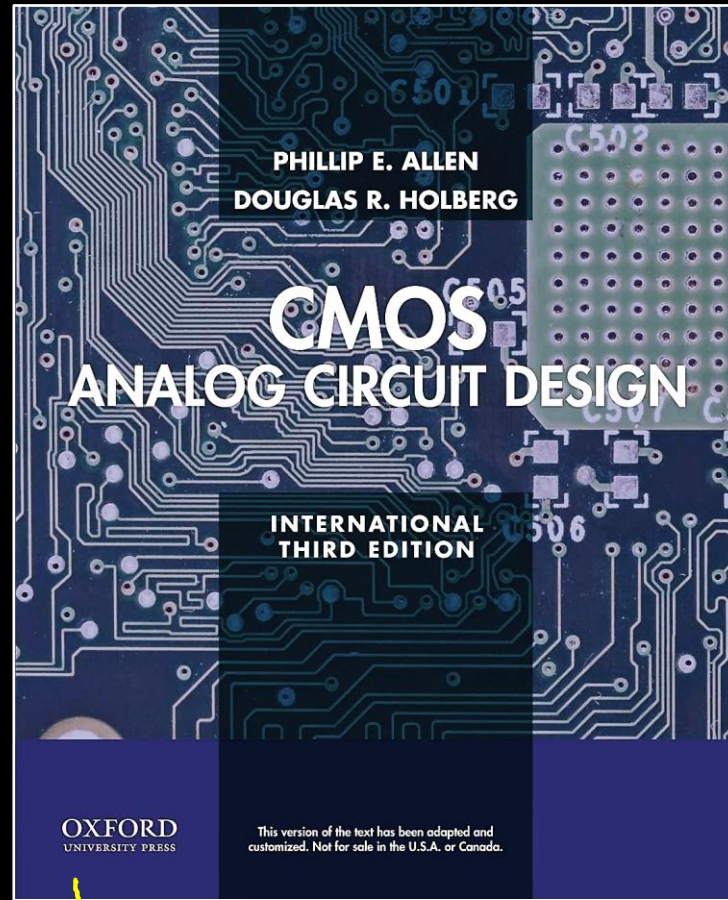


→ INDUSTRY →

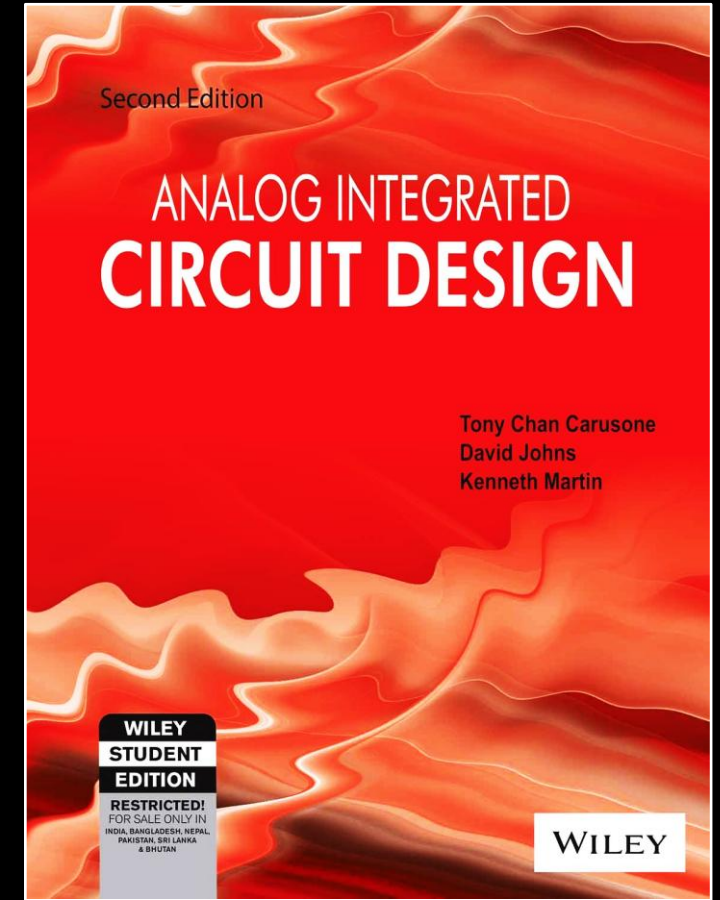
ANALOG DESIGN BOOKS



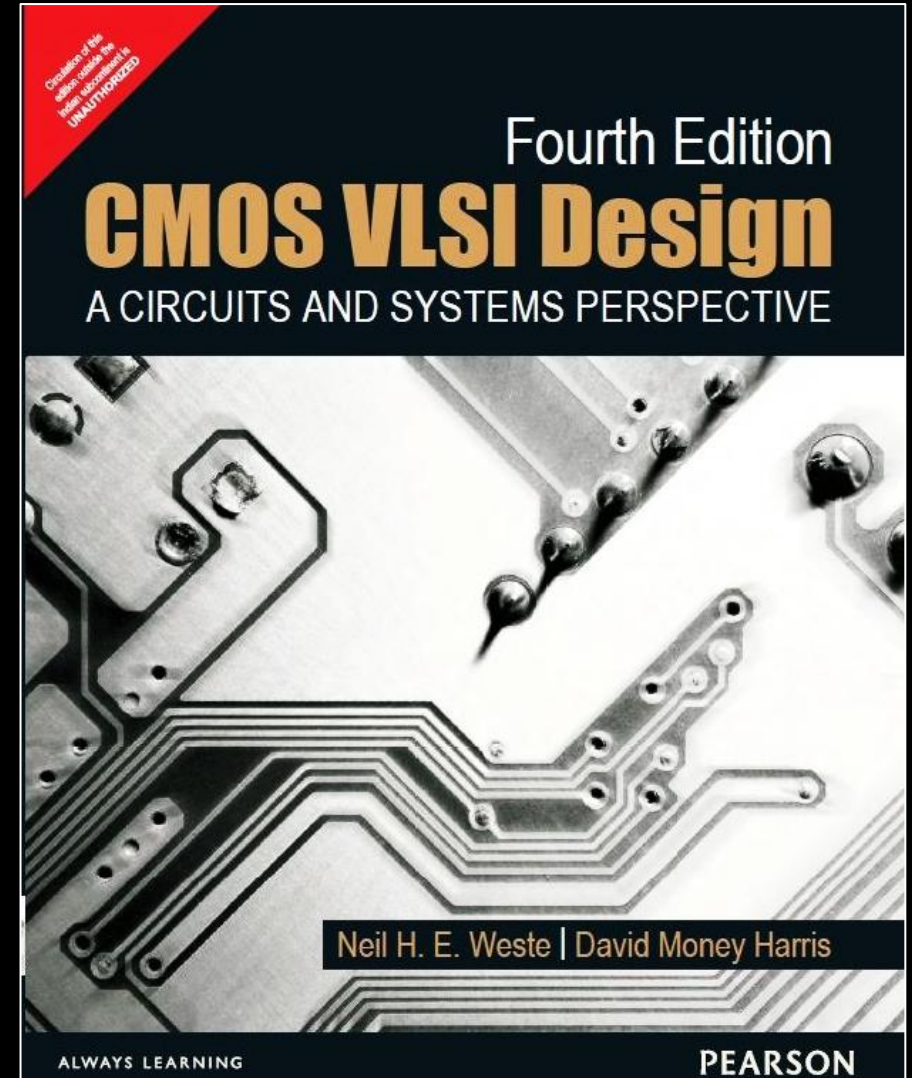
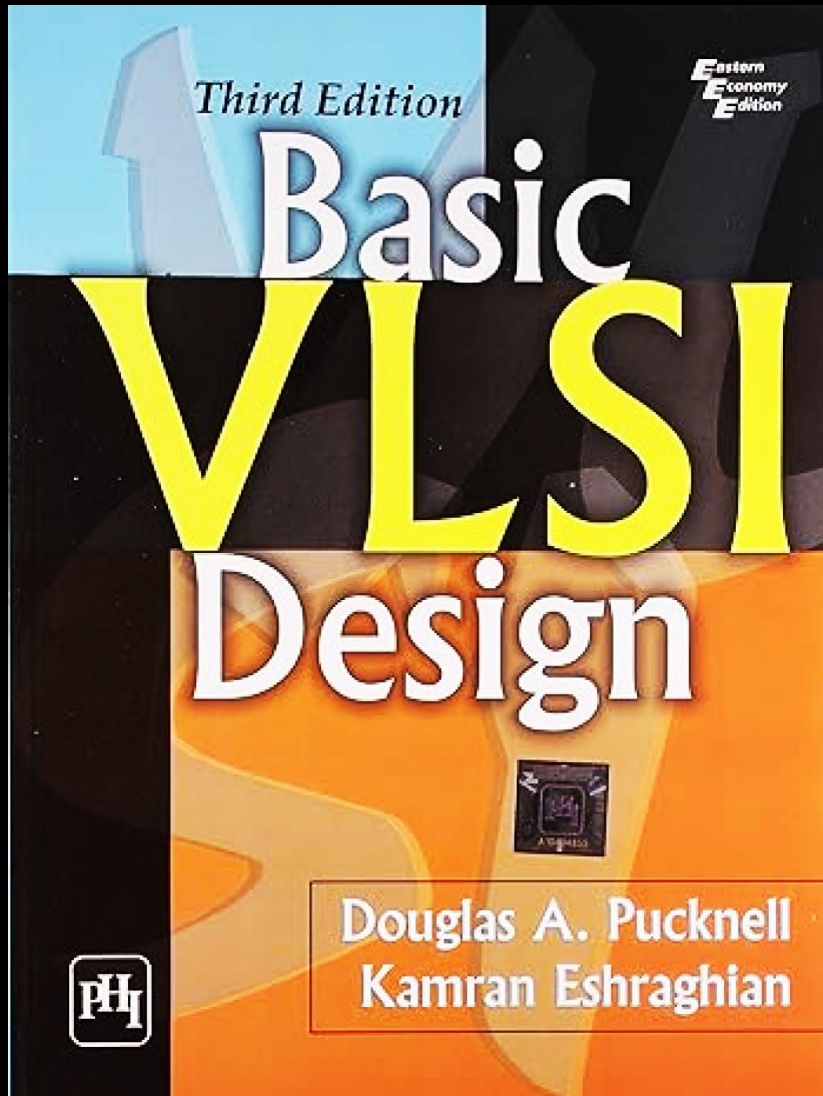
→ INDUSTRY PROJECTS .



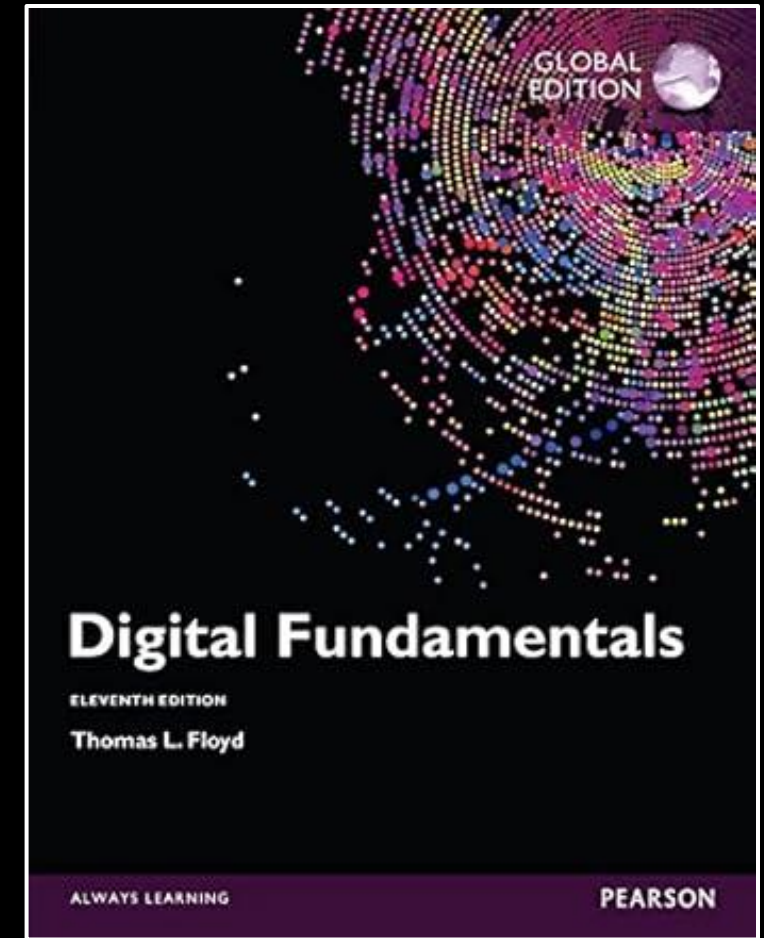
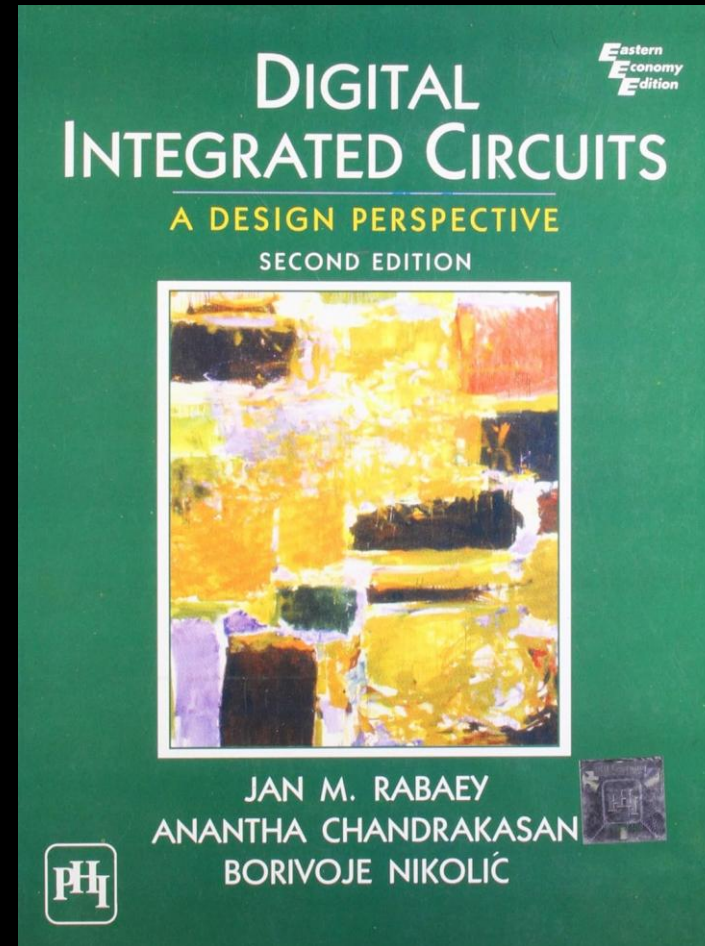
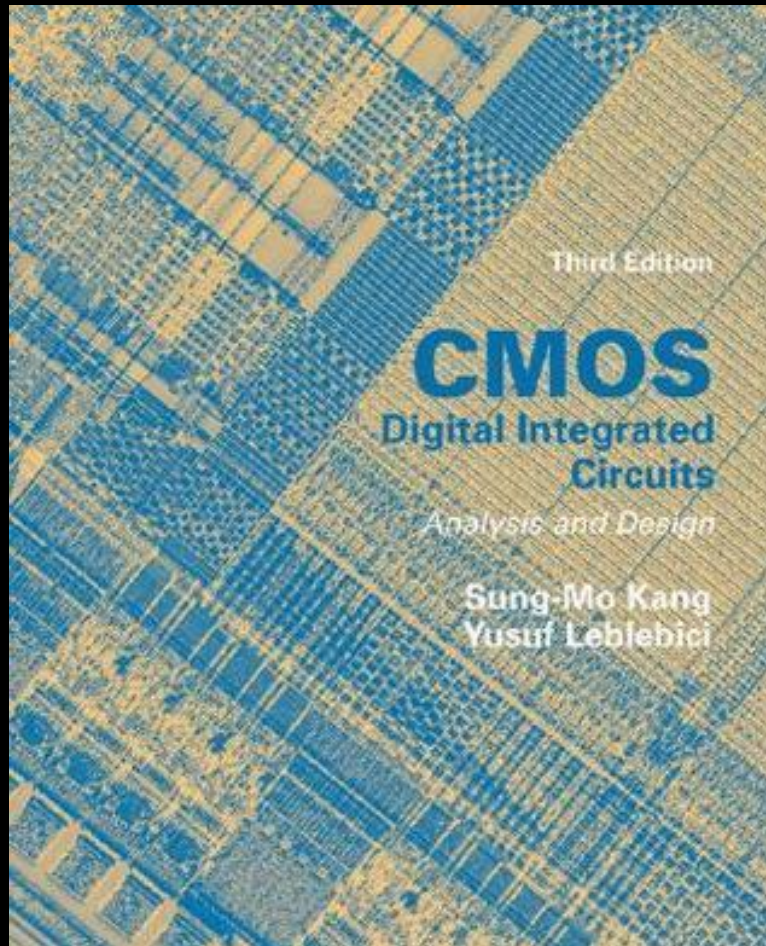
→ MOST Preferred ✓.



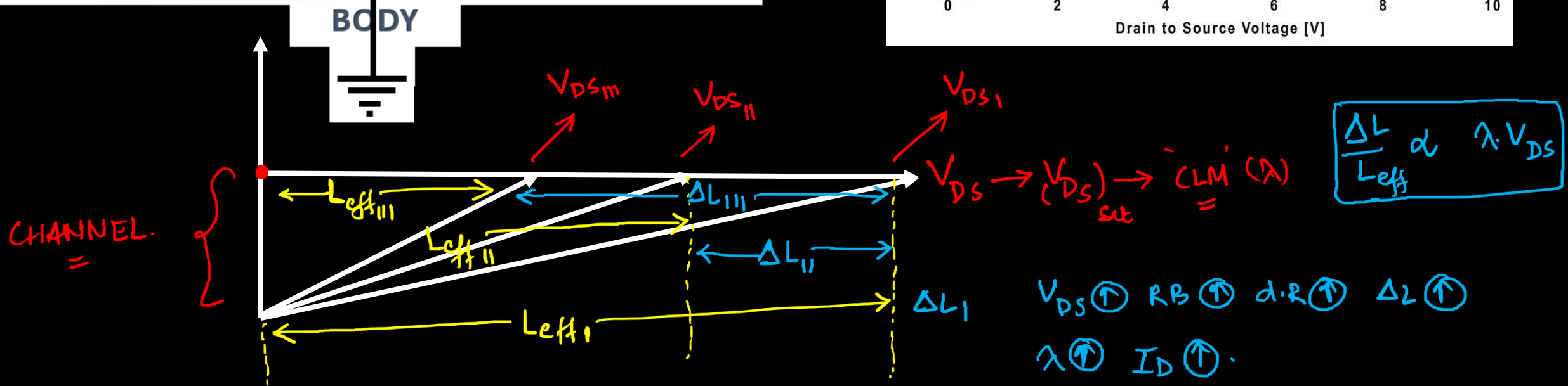
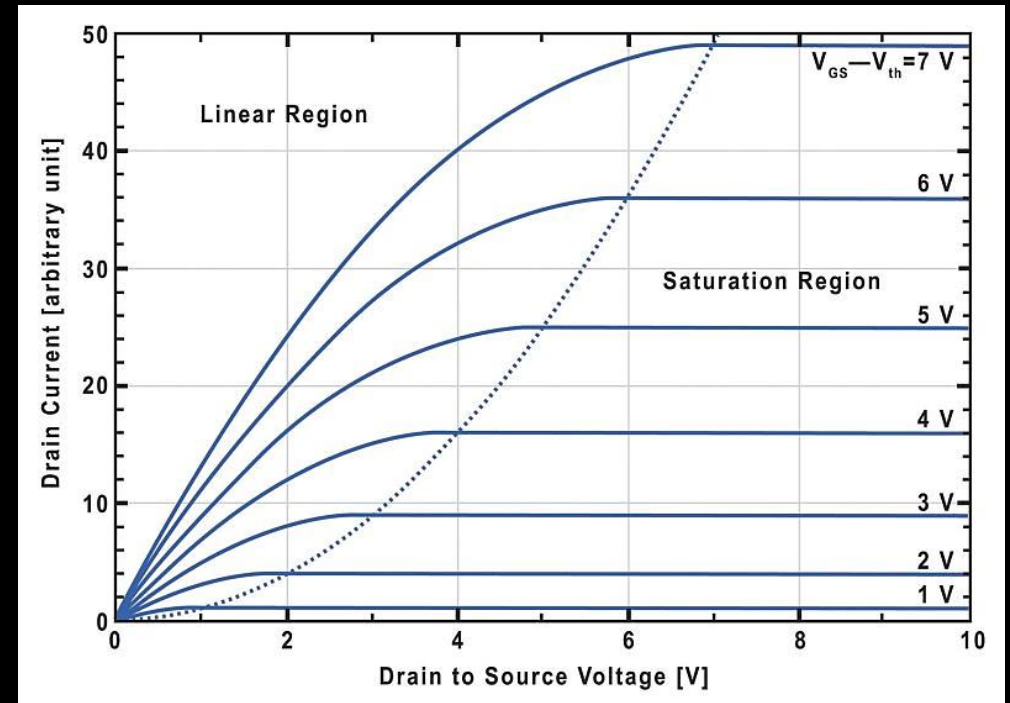
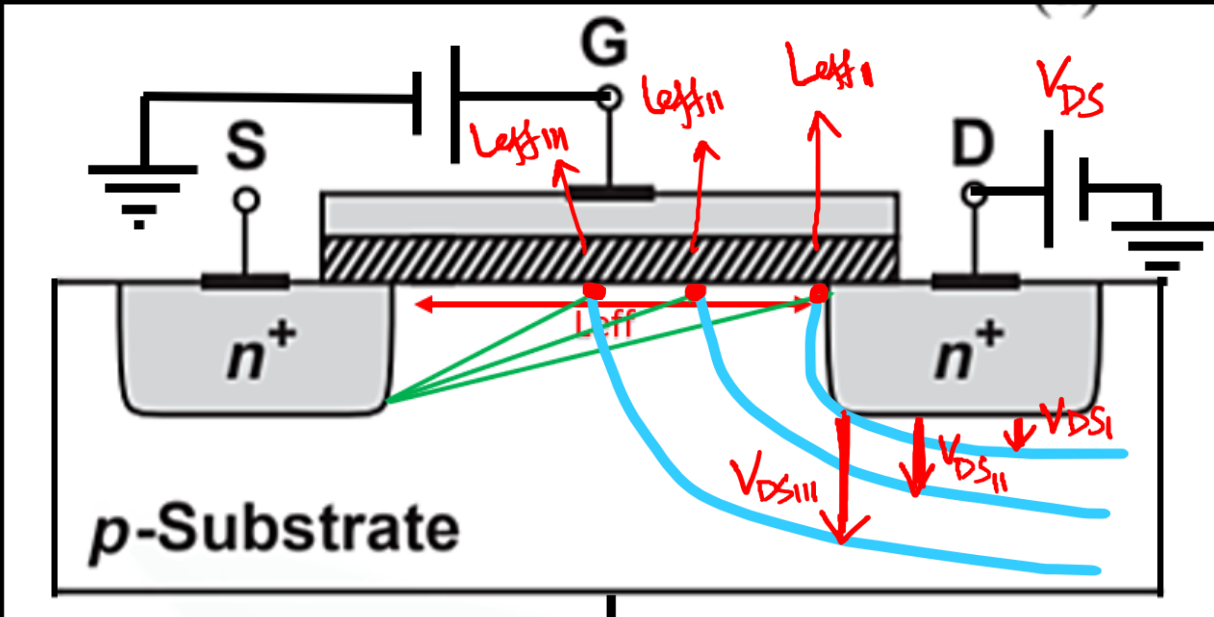
ANALOG LAYOUT & MISC. BOOK



DIGITAL VLSI THEORY BOOKS



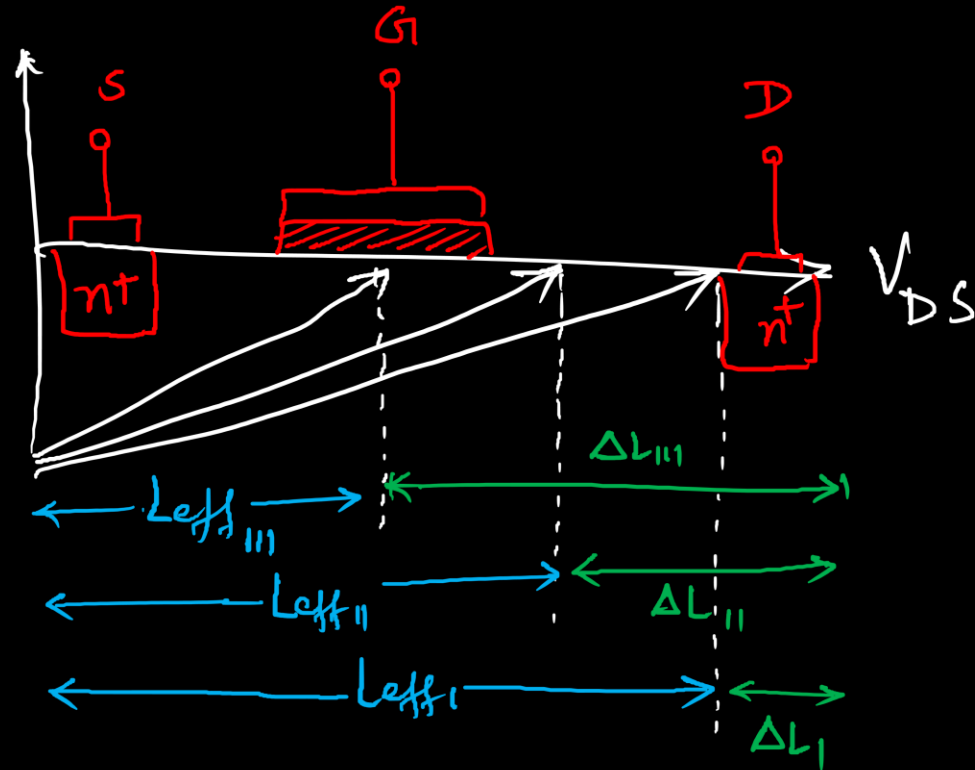
Concept of CHANNEL LENGTH MODULATION [CLM]



INTERVIEW :- State the Relationship Between L_{eff} and λ w.r.t different V_{DS} values in a graph.

$$\frac{\Delta L}{L_{eff}} \propto \lambda \cdot V_{DS} \propto (I_D)_{sat}$$

$$(I_D)_{sat} = \frac{1}{2} \mu_n C_{ox} \cdot \frac{W}{L} (V_{GS} - V_{th})^2 (1 + \lambda \cdot V_{DS})$$



- $\Delta L_I < \Delta L_{II} < \Delta L_{III}$
- $L_{effI} > L_{effII} > L_{effIII}$
- $V_{DS_{III}} > V_{DS_{II}} > V_{DS_I}$
- $I_{D_{S_{III}}} > I_{D_{S_{II}}} > I_{D_{S_I}}$

NOTE
#

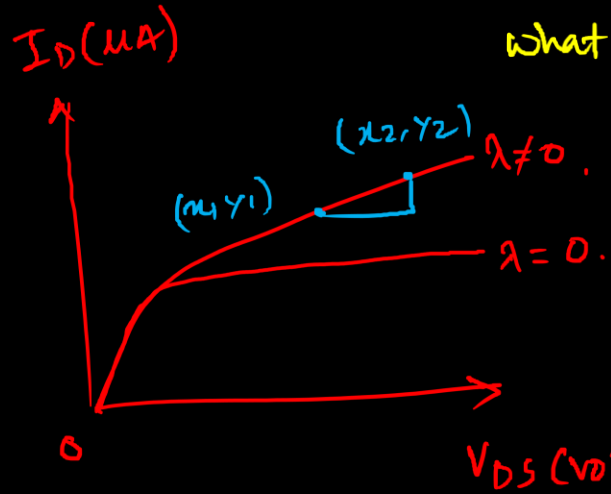
$$L_{eff} \propto \sqrt{V_{DS} - V_{eff} + \Phi_0}$$

$\downarrow$
 $= V_{ov}$
 $= V_{GS} - V_{th}$

Surface potential

"CLM" $\rightarrow$ is defined as the variation of channel in MOSFET when $V_{DS} \rightarrow V_{DS_{sat}}$ as a resultant of which the effective channel length reduces to an abnormal amount. It's a characteristic of a SHORT-CHANNEL device.

INTERVIEW :- What is the Relationship Between $(I_D)_{sat}$, V_{DS} and CLM coeff (λ).
OR'



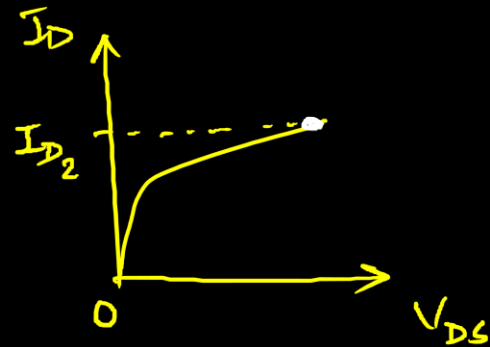
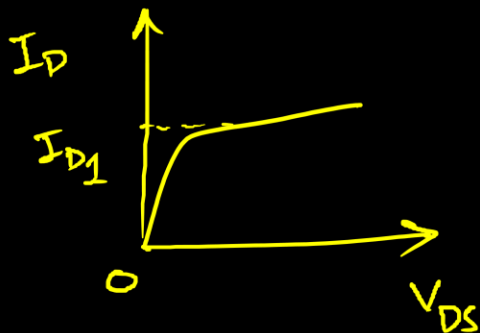
What is the Slope of I_D , V_{DS} graph?

$$\frac{(I_D)_I}{(I_D)_{II}} = \frac{1 + \lambda \cdot V_{DS_I}}{1 + \lambda \cdot V_{DS_{II}}}$$

$$\text{slope} = \frac{\partial I_D}{\partial V_{DS}} =$$

$$\frac{(I_D)_I}{1 + \lambda \cdot V_{DS_I}} = \frac{(I_D)_{II}}{1 + \lambda \cdot V_{DS_{II}}}$$

INTERVIEW :- You're given 2 graphs; write the Relationship Between their Effective channel lengths.



$$\frac{\Delta L}{L} \propto \lambda \cdot V_{DS} \propto (I_D)_{sat}$$

$$I_{D2} > I_{D1}$$

Hence, Device - 2 is a SHORT-CHANNEL DEVICE...

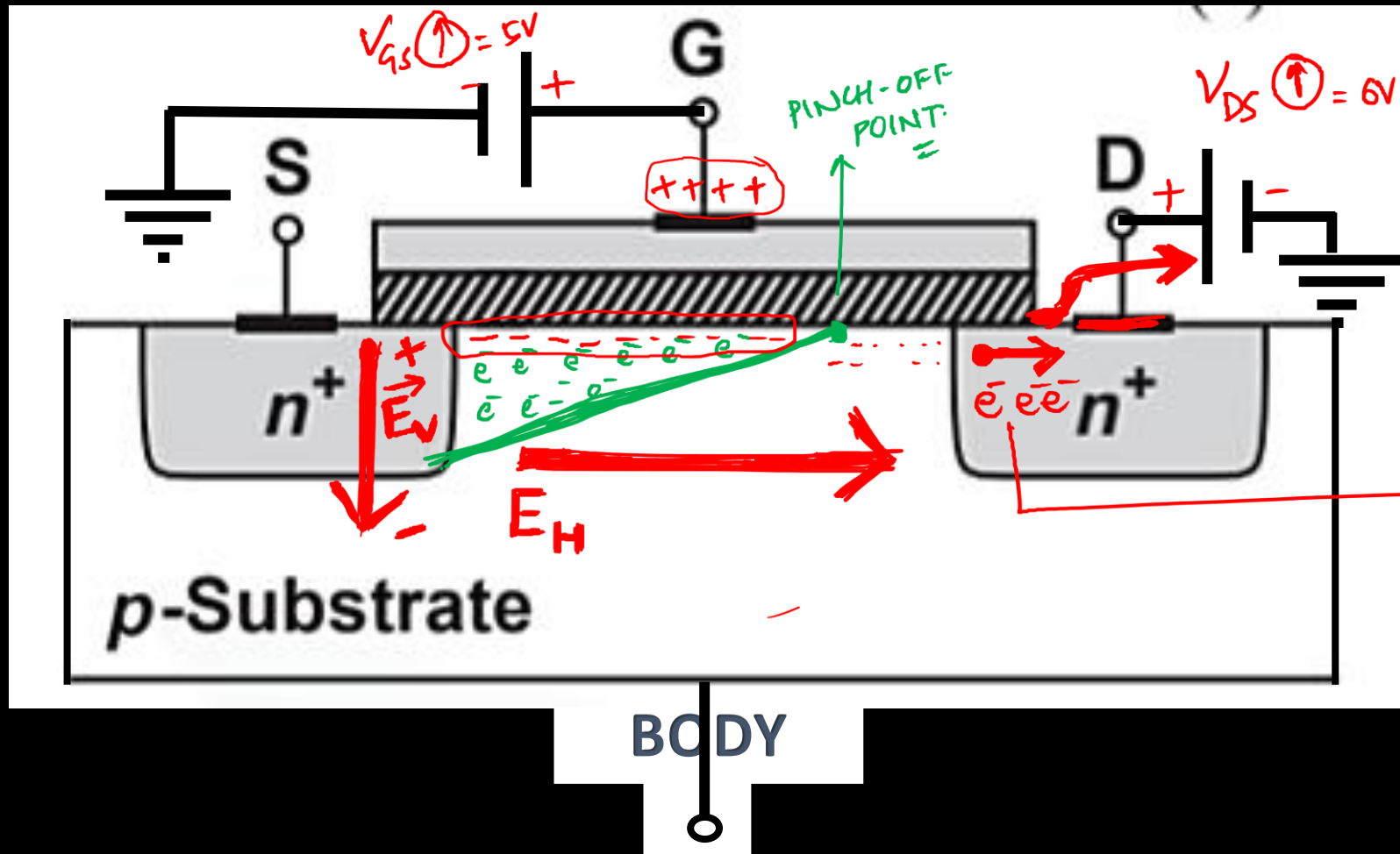
INTERVIEW :- What will happen to the MOSFET in the Absence of CLM ?

Solⁿ :-

$$(I_D)_{sat} = \frac{1}{2} \mu_n \cdot C_{ox} \cdot \frac{W}{L} \left\{ (V_{GS} - V_{th})^2 \cdot (1 + \cancel{\lambda} \cdot V_{DS}) \right\}^0$$

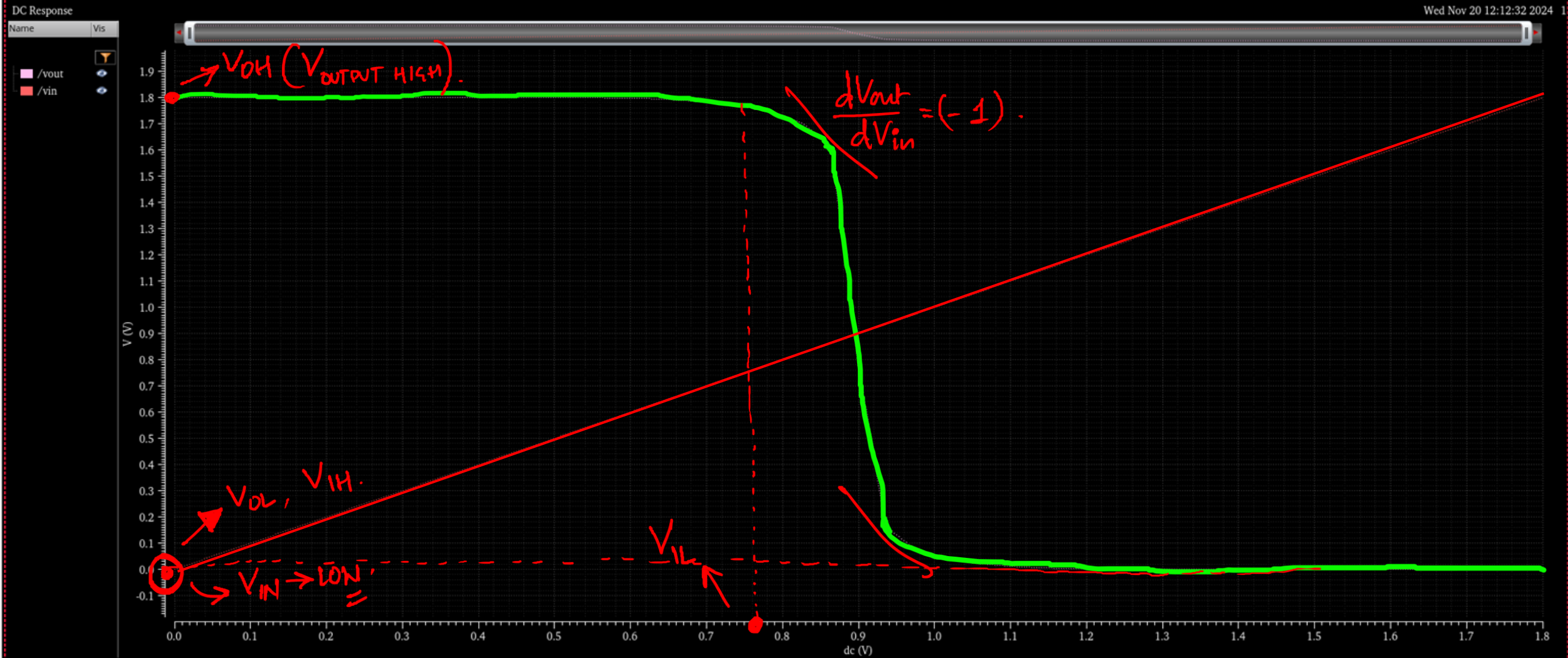
$$(I_D)_{sat} = \frac{1}{2} \mu_n \cdot C_{ox} \cdot \frac{W}{L} \left\{ V_{GS} - V_t \right\}^2$$

0 0 MOSFET's operation will be now Independent of V_{DS} .



In CM; V_{eff} decreases to very less distance from D' ; So
 → how come is it possible, we get a $\neq 0 I_D$?

→ Due to high $\vec{E}$ at the D' region; e^- s are pulled towards the Metal Contact inside the device and hence, we get $\neq 0 I_D$.



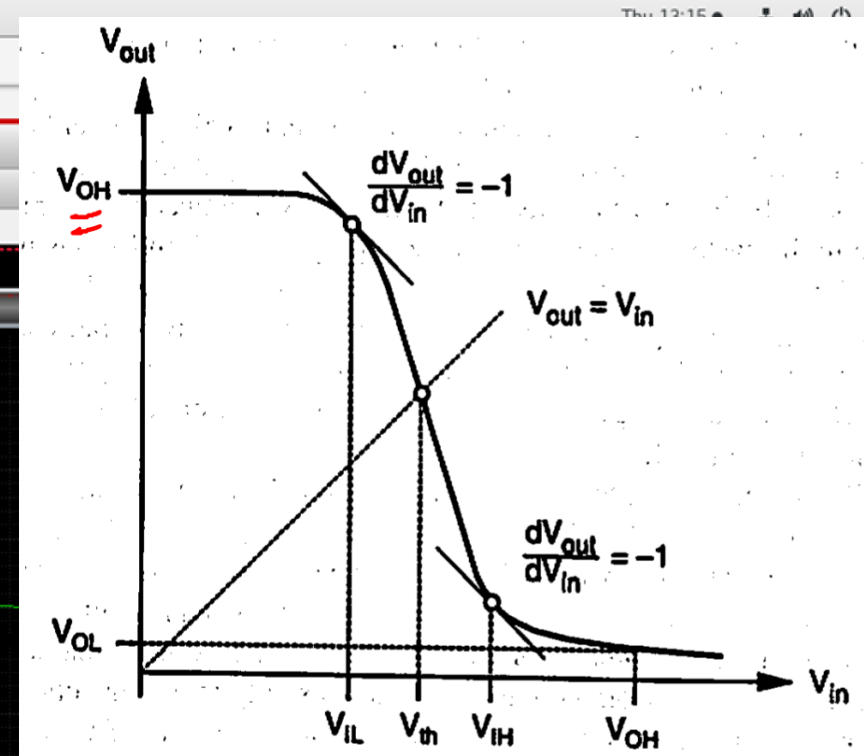
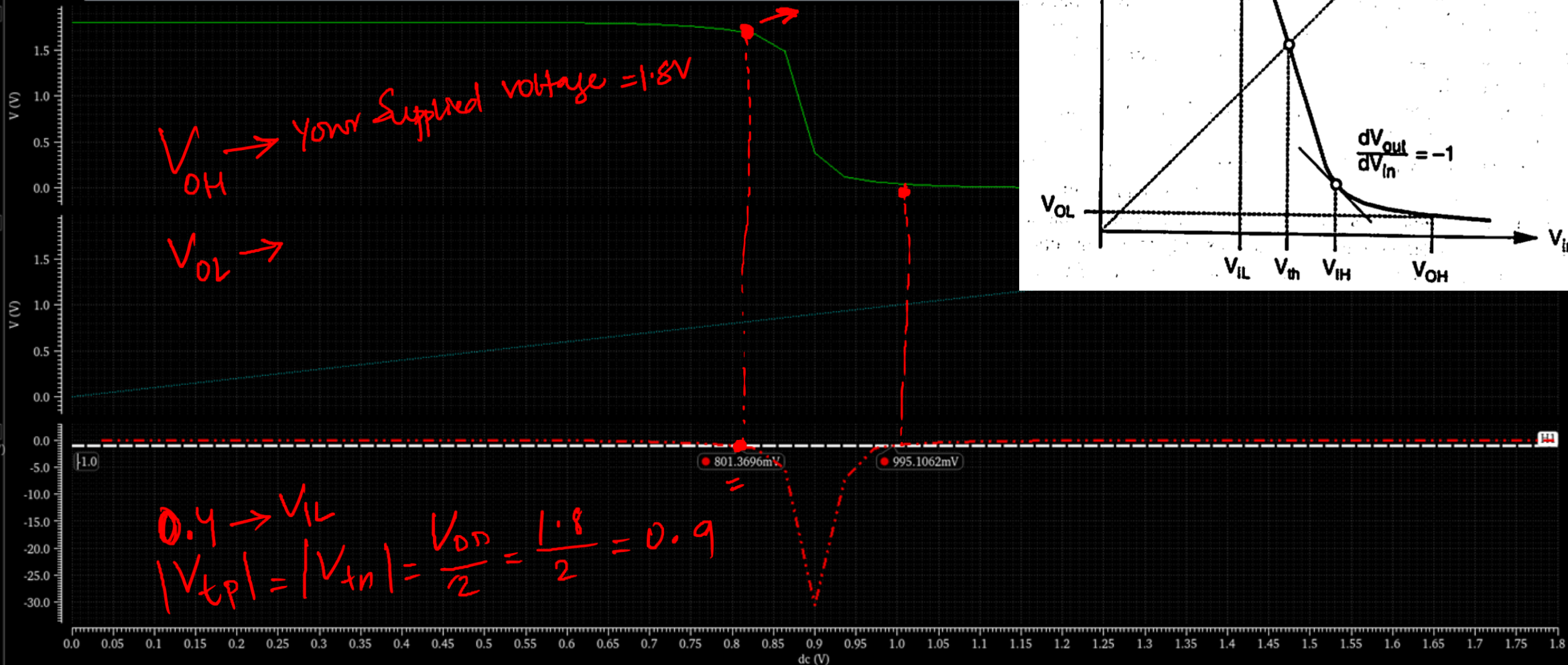
DC Response

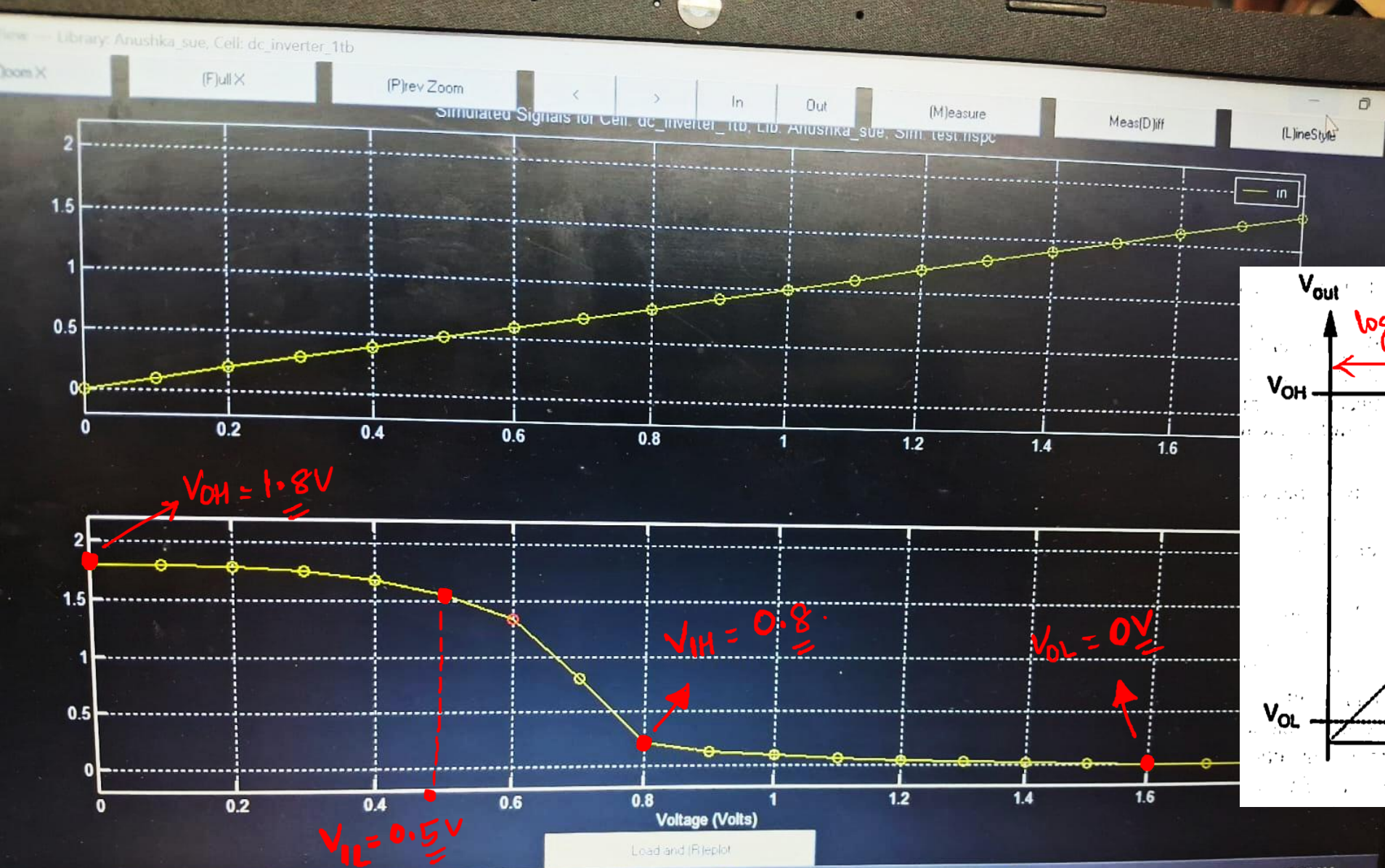
Name

/vout

/vin

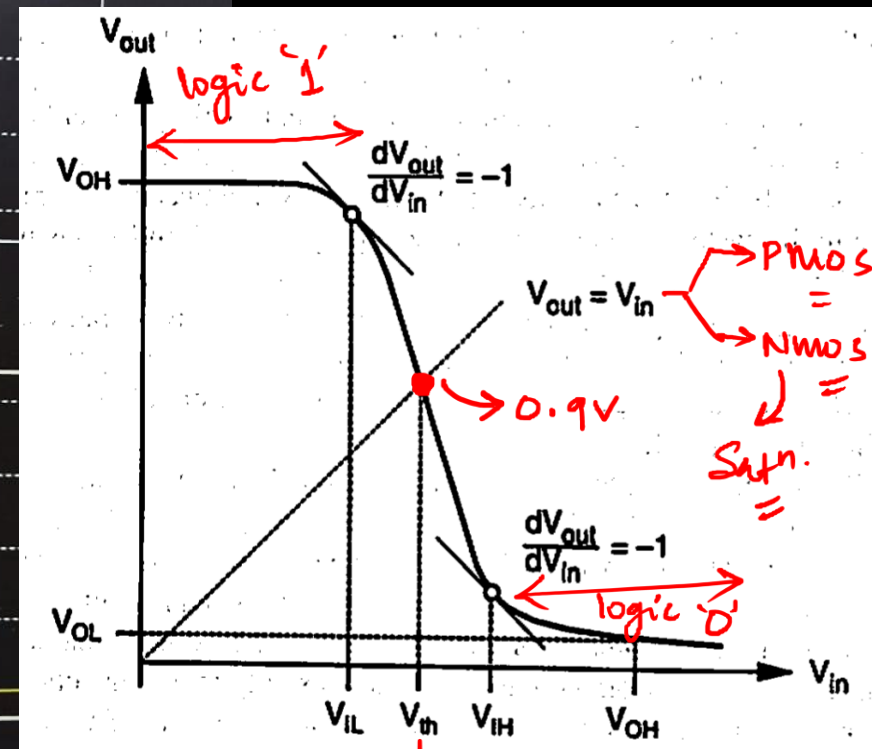
deriv(VS("/vout"))





$$|V_{tp}| = V_{tn} = \frac{V_{DD}}{2} = \frac{1.8V}{2}$$

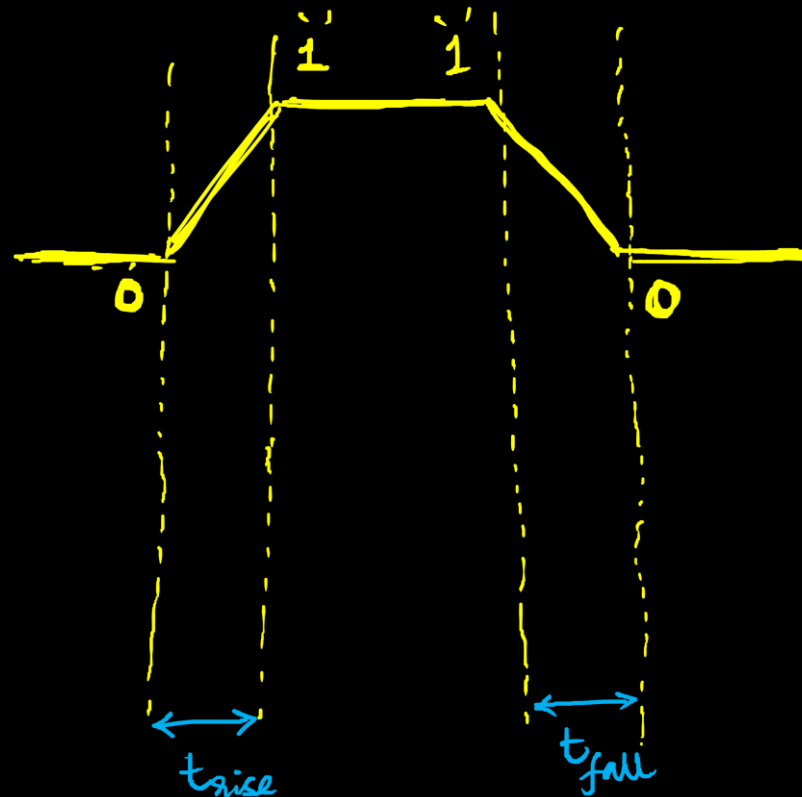
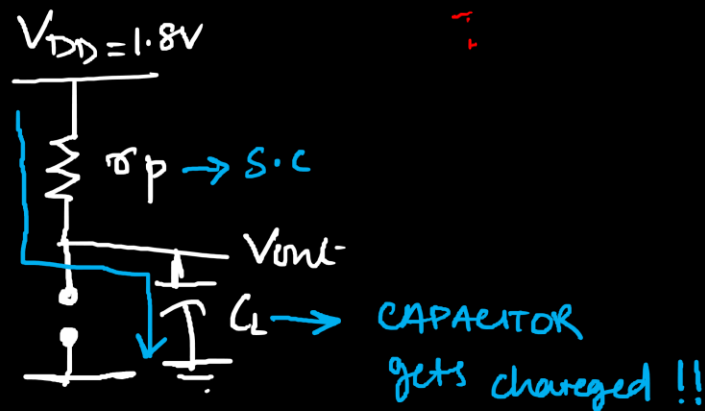
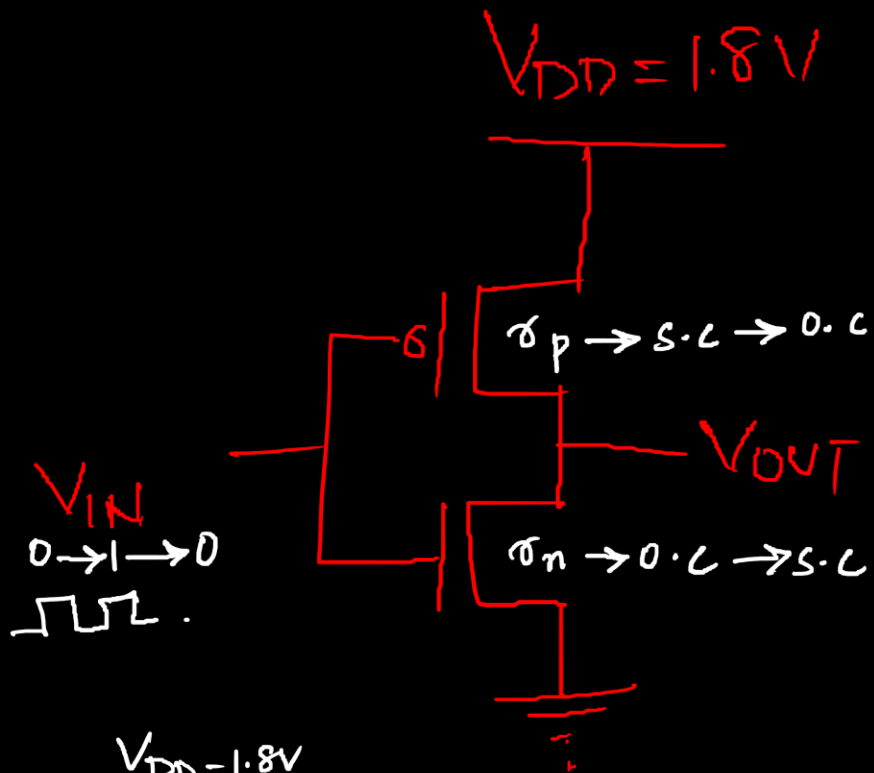
$$\rightarrow |V_{tp}| = V_{tn} = 0.9V$$



Switching threshold = $\frac{1}{2}$ of V_{supply} .

circuit = CMOS INVERTER

DYNAMIC - CHARACTERISTICS



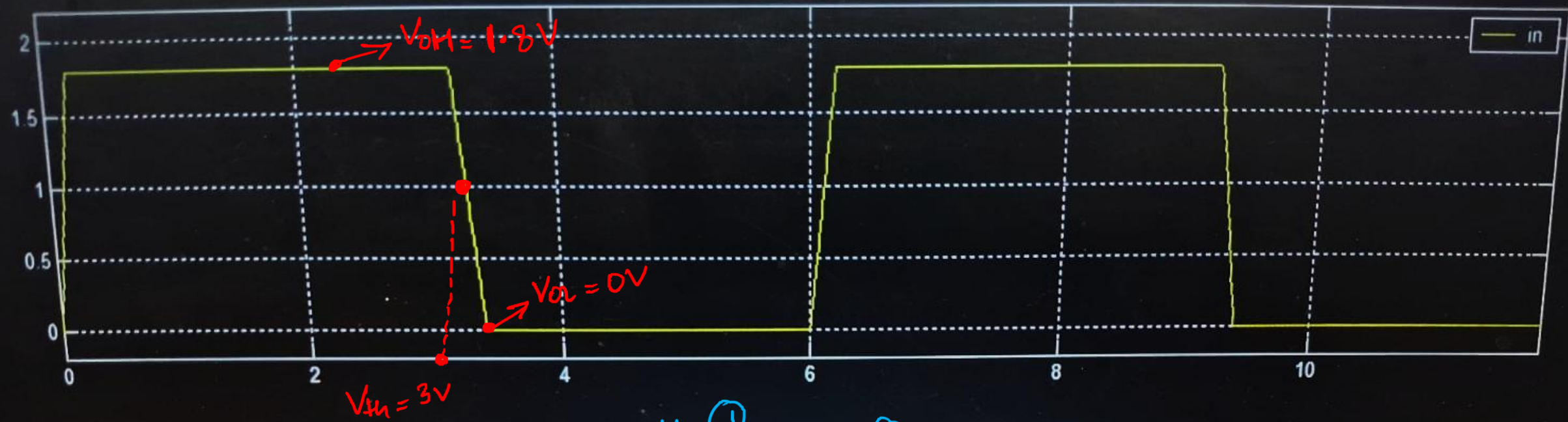
$$Z_{PLH}(0 \rightarrow 1) = 0.69 R_p \cdot C_L$$

$$Z_{PLH}(1 \rightarrow 0) = 0.69 R_n \cdot C_L$$

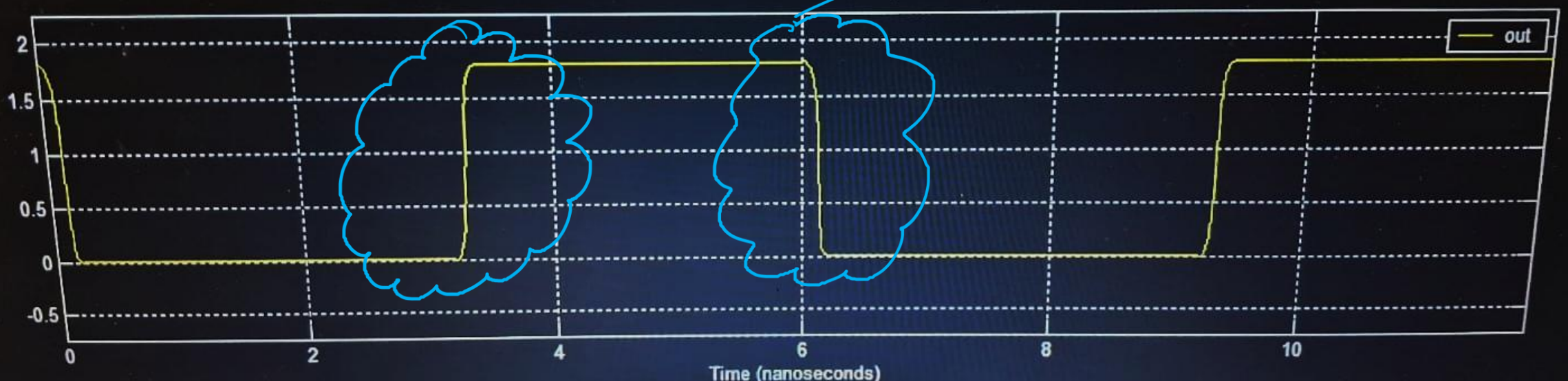
$$Z_{Pd} = \frac{Z_{rise} + Z_{fall}}{2}$$



Simulated Signals for Cell: inverter4_1_tb, Lib: Anushka_sue, Sim: test.nspc

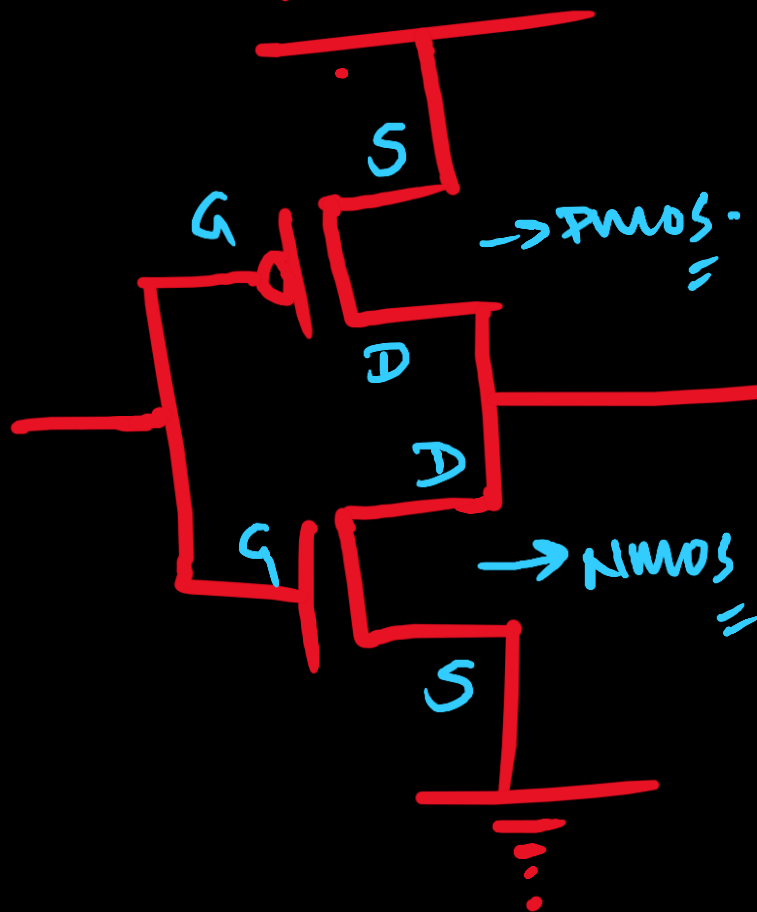


MAGNIFY !! ①



$$V_{DD} = 1.8V$$

$$V_{tp} = V_{tn} = 0.4V$$



$$\begin{aligned} V_{in} &= 0 \\ &= 1.8V \\ &= 1.6V \\ &= 0.9V \end{aligned}$$

V_{out}
1

$$\begin{aligned} 0V &= \\ 0.2 &= \\ 0.9 &= \end{aligned}$$



$V_{in} \uparrow \quad V_{out} \downarrow$